

- Azure Lighthouse allows **cross-tenant visibility and management**. To **correlate data across multiple tenants in Sentinel**, you must use Lighthouse to **grant delegated access** to workspaces in the other tenant.
- Microsoft Defender for Cloud Apps provides **file activity tracking**, including **downloads, deletions, accesses**, and can **generate activity logs** for suspicious behavior over specific timeframes.
- Safe Attachments in Defender for Office 365 can **automatically scan uploaded files to SharePoint, OneDrive, and Teams** for **malware** and **quarantine malicious files**.
- **IsNewCountry** is a built-in behavioral analytics flag that indicates **a user's first activity from a country they've never signed in from before**. Perfect for detecting **location-based anomalies**.
- Playbooks in Sentinel are **automated workflows** built with **Logic Apps**. You can trigger a playbook (e.g., to block IPs or isolate a VM) when an **Analytics Rule** detects a brute-force attack.
- **Fusion rules** in Sentinel use **machine learning to correlate signals across products**, reducing noise by surfacing **only meaningful, high-confidence incidents**.
- **Watchlists** store reference data (like **lists of bad IPs**) that can be used in **KQL queries, analytics rules, or hunting** to enrich incidents or filter results.
- Microsoft Defender for Cloud alerts are sent to the **SecurityAlert** table in Sentinel. Key Vault-related attacks, including brute-force attempts, would appear here.
- Microsoft Defender for Endpoint on iOS can detect and report usage of risky apps. Combined with **Microsoft Defender for Cloud Apps** (App Control), you can **block or restrict** unauthorized cloud app usage in real-time.
- To meet the requirement for **label tracking from Windows 10 devices**, you must first **enable advanced features** in **Microsoft Defender Security Center** to integrate MDE with AIP's **Data discovery dashboard**. (AIP -> (Azure Information Protection))
- Microsoft Defender for Endpoint integrates with **Microsoft Defender for Cloud Apps** (aka MCAS) to block access to **unsanctioned cloud apps**. This is done by **enforcing app control policies** via network filtering on endpoints.
- When creating an analytics rule in Sentinel, you can define **event grouping settings** (e.g., group by user account), so that **each test user generates a separate incident**.
- Defender for Cloud sends its alerts to the **SecurityAlert** table, regardless of whether the underlying OS is **Windows or Linux**.
- To manage and attach playbooks (Logic Apps) to Sentinel rules, the user must have the **Microsoft Sentinel Automation Contributor** role.

- To create a custom query-based analytics rule in Sentinel that automatically runs a playbook -> You create a **custom KQL-based analytics rule**, then attach a **Logic App (playbook)** to trigger automatically when that rule fires.
- Defender for Identity (formerly Azure ATP) is designed to **monitor domain controllers** by installing a **sensor directly on the DC**, enabling detection of identity-based threats.
- To monitor **download behavior from SharePoint Online** (especially before a user is deleted), use a **file policy in Microsoft Defender for Cloud Apps**.
- To **enable Microsoft Defender for Cloud protection on on-premises Linux servers**, the **first step** is to install the **Log Analytics agent**.
- **Azure Arc** is a Microsoft service that lets you **manage and secure infrastructure and services outside of Azure**
To suppress specific Defender for Cloud security alerts at the root management group level with minimal administrative effort, you should **create an Azure Policy assignment**.
- To protect on-premises computers with **Microsoft Defender for Cloud**, the **Log Analytics agent must be installed and connected to a workspace**.
- To view **who modified sensitivity labels, when, and how**, use **Activity explorer** in the **Microsoft 365 Compliance Center**.
- **Zero-Hour Auto Purge (ZAP)** is a **Microsoft Defender for Office 365** feature that **automatically removes malicious emails** from user inboxes **after delivery**, if they're later found to be threats, ZAP is enabled by default.
- **Microsoft Exchange Online** is Microsoft's **cloud-based email and calendaring service**, part of Microsoft 365.
- To check if **ZAP moved a message after delivery**, use the **Threat Protection Status report** in **Microsoft Defender for Office 365**. It's the only report that logs **ZAP actions** directly.
- Two possible causes for the **intermittent** failures of your custom analytics rule in Azure Sentinel are:
A. The rule query takes too long to run and times out. D. There are connectivity issues between the data sources and Log Analytics.
- Brute force attack against Azure Portal analytics rule has been triggered. You need to identify the geolocation information that corresponds to the incident. -> Potential malicious events:
When traffic is detected from sources that are known to be malicious, Microsoft Sentinel alerts you **on the map** (Potential malicious events map). If you see orange, it is inbound traffic. If u see Red its outbound.

- **msticpy** is a library for InfoSec investigation and hunting in Jupyter Notebooks. It includes functionality to: query log data from multiple sources. enrich the data with Threat Intelligence, geolocations and Azure resource data. extract Indicators of Activity (IoA) from logs and unpack encoded data
- **MSTICPy** reduces the amount of code that customers need to write for Microsoft Sentinel, and provides: - **Data query capabilities**, against Microsoft Sentinel tables, Microsoft Defender for Endpoint, Splunk, and other data sources. - **Threat intelligence lookups with TI providers**, such as VirusTotal and AlienVault OTX. **Enrichment functions** like geolocation of IP addresses, Indicator of Compromise (IoC) extraction, and WhoIs lookups. - **Visualization tools using event timelines**, process trees, and geo mapping. Advanced analyses, such as time series decomposition, anomaly detection, and clustering.
- The Evidence and Response tab of Microsoft 365 defender portal shows all the **supported events** and **suspicious entities** in the alerts in the incident.
- **Fog data** refers to data processed and analyzed **closer to the source** (like IoT devices or edge systems), rather than sending everything to a centralized cloud or data center.
- To monitor 100 Linux virtual machines in your Azure subscription using Microsoft Sentinel while minimizing administrative effort and parsing required to read fog data, you should configure a **Common Event Format (CEF) connector**, which sends standardized and easily readable data from a CEF server listening for Syslog from all the Linux VMs to Sentinel.
- **Azure Event Hub** is a **big data streaming platform and event ingestion service** from Microsoft Azure. It is used to **collect, process, and store large volumes of real-time data** (events) from various sources. ->
- To stream Defender for Cloud logs **efficiently to a Syslog server**, send them to **Azure Event Hub**, and manage the setup **centrally using Azure Policy** at the **root management group** level.
- To stream alerts into //Syslog servers// ,and other monitoring solutions, connect Defender for Cloud using continuous export and Azure Event Hubs.
- To stream alerts at the tenant level, use this //Azure policy// and set the scope at the root management group.
- **Auto-provisioning** is a feature in **Microsoft Defender for Cloud** that automatically installs the required **monitoring agents** (like the **Log Analytics agent**) on **new Azure resources** — so they can start being **monitored and protected immediately**, without manual setup.
- A machine with Azure Arc-enabled servers becomes an Azure resource (*even though is a Linux server hosted on AWS*) and - when you've installed the Log Analytics agent on it - appears in Defender for Cloud with recommendations like your other Azure resources. Install Log Analytics manually or when you enable auto provisioning of Log Analytics in "Autoprovisioning" tag, auto provisioning is already turned on.
- To **automatically detect and label sensitive documents** in SharePoint Online, you should use **Azure Information Protection** — which integrates directly with DLP and Microsoft 365 compliance tools.
- RegEx can be used in Microsoft 365 to create custom sensitive information types that detect unique data patterns—like 32-character account numbers—for use in DLP and auto-labeling policies. (*in case there is no any build any sensitive pattern types that supports our case*)

- Visual Basic for Applications (VBA) is a computer programming language developed by Microsoft. You can create macros to automate repetitive word- and data-processing functions with VBA and generate custom forms, graphs, and reports.
- To safely hide false positives in Microsoft Defender for Endpoint, **hide the alert, create a scoped suppression rule, and allow alerts to still generate for auditing.**
- To **generate an alert from your query** that includes full context, you **must include DeviceId and ReportId in the project clause** of **your query**.
- To perform temporary, automated actions across all devices, **create a custom device group with a tag filter**, tag all target devices, and **promote the new group to the highest rank** so it overrides existing group membership.
- Honeytoken Accounts: In the context of security, a honeypot is typically a decoy or a fake account set up to lure attackers. These accounts are monitored closely, and any access attempts to them trigger alerts to security teams, indicating potential malicious activity.
- **Sensitivity tags** are meant for active users and groups.
- **Dynamic Delivery** in Microsoft Defender for Office 365 sends emails right away, but temporarily replaces attachments with placeholders until they're safely scanned. Enable to preview attachments in a **safe mode** while scanning is taking place. This avoids email delays and still protects users from malicious files.
- **Top 20 by timestamp**, query returns the latest (recent) results by the timestamp.
- Only **EmailEvents** table contain the subject of the email.
- **Compliance Data Administrator** in Azure AD: This role provides permissions to manage data compliance settings and review compliance data. It includes capabilities like managing compliance features.
- **Security Administrator role** in Azure AD: This role has broad permissions across Microsoft 365 security services, including managing security policies, viewing security reports, and responding to security alerts. It can make significant changes to security settings across the organization.
- **Active Remediations Actions Role**: Approve and Reject pending actions generated by defender for endpoint.
- **Automatic enrichment** in the context of Microsoft Cloud App Security (MCAS) refers to the process of automatically augmenting security alerts and event data with additional contextual information. This helps in better understanding and analyzing the security events, improving the accuracy of threat detection, and reducing false positives. (Geolocation data, IP address, user agent strings, Threat Intelligence and Cloud app discovery)
- **Zero-Hour Auto Purge (ZAP)** is an automated email security feature that continuously monitors and assesses the threat landscape to identify emails that were initially delivered to mailboxes but later determined to be malicious. ZAP helps ensure that such threats are swiftly neutralized, minimizing potential harm to the organization.
- The **Threat Protection Status Report** shows the **number of emails with malicious content** (e.g., harmful files or URLs) that were **blocked** by features like the **anti-malware engine, ZAP, Safe Links, Safe Attachments, and anti-phishing policies** in **Microsoft Defender for Office 365**.

- **Contributor** role has **modification privileges**, such as applying/remediating recommendations — which are beyond read-only access (security Reader role)
- Only the **Owner** role has **full management rights at the subscription level**, including assigning policies (initiatives).
- Labeling activities are available in Activity Explorer (ex: changes made to sensitive labels etc..)
- The **Modifications of sensitive groups** report in **Microsoft Defender for Identity** helps track all changes to high-privilege groups like **Domain Admins** within the past 30 days, enabling detection of potential security threats in Azure AD.
- A **hybrid** Azure Active Directory (Azure AD) tenant is a setup where an organization integrates its on-premises Active Directory (AD) environment with Azure Active Directory.
- **Azure AD Connect** is the tool that facilitates the synchronization of on-premises AD objects (users, groups, contacts) to Azure AD. It ensures that changes made on-premises are reflected in Azure AD and vice versa.
- **Single Sign-On (SSO)**: Users can sign into cloud resources using their on-premises credentials, providing a seamless login experience.
- **Aggregated alerts** are **grouped alerts** that combine multiple related security events into a **single alert instance** to reduce noise and simplify investigation.
- To investigate an aggregated alert in Microsoft 365, use the **"Events" tab** in the DLP alert dashboard. It shows all triggering events and impacted entities (e.g., files, emails), helping you identify the user, device, and action involved.
- To **create a custom tracked query** for assessing threats in **Microsoft 365 Defender**, you should use the **Advanced Hunting** page.
- To minimize admin effort and cover the full IP range properly, **prepare an import file with each IP**, then import it via **Microsoft 365 Defender portal**. **(cannot Use CIDR (/27) — Defender does not support CIDR import).**
- **Review security recommendations and enable server vulnerability scans**: This task involves reviewing security recommendations and enabling scans. The Security reader role is suitable for reviewing recommendations, but enabling scans might require higher privileges. Therefore, the Security administrator role is also appropriate for this task.
- **join kind=inner**: Efficient join method — only returns matching rows from both tables.
- **Kerberos Delegation**: Kerberos is an authentication protocol used in many on-premises networks. Delegation allows a service to impersonate a user to access resources on their behalf.
- **LDAP** (Lightweight Directory Access Protocol) is a protocol used to access and manage directory information over a network. LDAP is widely used in enterprise environments to query and modify directory services such as Microsoft Active Directory.
- To remediate the unsecure Kerberos delegation risk in Microsoft Defender for Identity, modify the delegation settings of the exposed computer objects in Active Directory.
- **Delegation settings** control whether and how a computer or service can **impersonate users** to access resources on their behalf — typically using **Kerberos authentication**.
- To remove a file from quarantine after an automated investigation, use the **History tab in the Action center** to **revert the action** in Microsoft 365 Defender.

- Use the **Threat Analytics** blade to stay informed on new attack techniques and understand their impact on your organization's resources.
- **Pass-through Authentication (PTA)** allows users to sign into both on-premises and cloud based applications using the same credentials. When a user tries to sign into an Azure AD integrated application, the authentication request is passed through to the on-premises AD to validate the user credentials.
- **Password hash synchronization (PHS)** is another feature of Azure AD Connect that synchronizes the hash of the users' passwords from the on-premises AD to Azure AD. When PHS is enabled, Azure AD can authenticate users without needing to contact the onpremises AD.
- Reviewing the **certainty of the source computer** helps quickly filter out false positives and focus on genuine threats in Microsoft Defender for Identity.
- **LDAP** is a protocol used to access and manage directory services over a network. It is commonly used with Active Directory Domain Services (AD DS) to query and modify directory information.
 - To identify LDAP requests by AD DS users to enumerate AD DS objects, you need to use the "IdentityQueryEvent" table, which contains information about LDAP queries performed by users or applications.
- Use the **IdentityQueryEvents** table when working with **AccountSid** data related to **LDAP requests** in Microsoft Defender.
- The **Unified Audit Log (UAL)** in Azure is a comprehensive logging feature that allows administrators to track and monitor various activities and events across different Microsoft 365 services. This allows administrators to perform unified searches and gain insights into user and admin activities across their Microsoft 365 environment.
- **User Enrichment** in Defender for Cloud Apps adds external data to user profiles but is **not used for investigating threats** via audit logs.
- The **Azure connector** links Defender for Cloud Apps with **Azure services**, but doesn't help with **audit log investigations**.
- The **Office 365 connector** is essential as it enables Defender for Cloud Apps to **collect and analyze audit logs**, which is **crucial for threat investigation and response**.
- The columns NetworkMessageId and RecipientEmailAddress must be present in the output result of a query to apply actions to email messages. If NetworkMessageID is not mentioned in the summarize of a query, it won't work.
- **Agentless scanning** is a method of **assessing and monitoring systems** (like virtual machines, cloud workloads, or endpoints) **without installing a software agent** on the target machine. It's a way to **scan for vulnerabilities, configuration issues, or threats** from the outside — without deploying anything inside the system being scanned.
- In Microsoft Defender for Servers (Plan 1 or 2), when **agentless scanning is enabled**, and you want to **exclude a specific server like Server1**, the **most efficient and least administrative** method is to: **Assign an exclusion tag** to that server.
- **Interactive logon** is a **direct login by a user**, like from a console or RDP.
- **Non-interactive logon** happens **in the background**, typically by services or over the network (no user interactions)

- Use the table **IdentityLogonEvents** => for the interactive singings
- If you need a query to initiate device isolation for all affected devices. Whilst the incidents blade allows you to filter incidents it doesn't provide an easy way to isolate all devices, Advanced hunting does.

Terms

Microsoft Purview is a comprehensive set of data governance, compliance, and risk management solutions provided by Microsoft.

- 1. **Activities:** These are the actions performed in your environment that are logged for auditing purposes. They can include a variety of actions such as file sharing, downloading files, copying files, etc.
- 2. **Record Type:** This refers to the type of data source **from** which the activity records are retrieved. It can be Microsoft Teams, OneDrive, PowerBiAudit.
- 3. **Workload:** This refers to the specific service or **application where the activity took place**. It can be Microsoft Teams, OneDrive, PowerBi, SharePoint, etc. (from which application they shared the files)
- **Enabling automated investigation** allows for the use of Live Response capabilities, which includes initiating remote shell connections to onboarded devices. This setting is found under the advanced features in the endpoint settings of Microsoft 365 Defender.
- Microsoft **Teams chat messages** are stored in **users' Exchange Online mailboxes** (not in Teams itself). So: To **search Teams chats**, you must target **Exchange mailboxes**.
- Use the **Kind** keyword filter to limit results to specific message types like im (instant message), which covers **Teams chats**
- To search Teams chats in Microsoft Purview, select **Exchange mailboxes** and use the **Kind** keyword to filter for Teams messages.
- To **initiate the collection of investigation packages** from the Linux devices onboarded to Microsoft Defender for Endpoint, you should use: -> **Collect investigation package**: This action allows you to download an investigation package (ZIP file) containing relevant data and evidence related to the alert or incident. By selecting this option, you can gather the necessary information for further analysis and response.

In Azure security center, when you click on following things of an alert

- (Mitigate the threat) - this provides manual remediation steps for the present security alert (Provides recommendations on what to do to resolve the alert)
- (Prevent future attacks) - this provides security recommendations to help reduce the attack surface (future attacks), increase security posture, and thus prevent future attacks.(Provides recommendations for security in general such as Remediate Vulnerabilities and win defender should be activated on all devices)
- **Azure Defender for Key Vault** protects your secrets and keys by monitoring access patterns and alerting on suspicious activity — helping you prevent unauthorized access or misuse.
- To quickly reduce risk from suspicious IP access to Key Vault, **enable the key vault firewall** and restrict access to trusted IP ranges during investigation.

- The **Contributor** role will allow SecAdmin1 to perform tasks such as deploying resources and modifying resource properties within RG1, but it will not grant them access to perform administrative tasks at the subscription level. This will allow SecAdmin1 to apply quick fixes to the virtual machines using Azure Defender, while still adhering to the principle of least privilege.
- Microsoft Defender for Cloud (formerly Azure Security Center and Azure Defender) allows configuring email notifications by navigating to: **Environment settings → Select subscription → Email notifications.**
- The LAQueryLogs table provides details about log queries run in Log Analytics. Since Log Analytics is used as Microsoft Sentinel's underlying data store, you can configure your system to collect LAQueryLogs data in your Microsoft Sentinel workspace.

LAQueryLogs contains the details such as;

When queries were run

Who ran queries in Log Analytics

What tool was used to run queries in Log Analytics, such as Microsoft Sentinel

The query texts themselves

Performance data on each query run

2. The "FG" command in a live response session is used to bring a background process to the foreground. This allows you to interact with the process directly, which can be useful for monitoring or troubleshooting

3. To ingest Google Cloud Platform (GCP) data into Azure Defender

Configure the GCP security command centre -> Enable Security Health Analytics -> Enable the GCP security Command centre API -> Create a dedicated services account and a private key -> From azure security centre, add cloud connectors

References : <https://www.certbus.com/sc-200-questions.html>

4. Only Prefetch files are collected in investigation packages collected in live response sessions. Network transactions, Process History and Command History are not collected.

5. Investigation packages collected in live response sessions consists of -> Autoruns, installed programs, Network Connections, Prefetch files, Prefetch folder, Processes, Scheduled tasks, Security event log, Services, Windows Server Message Block (SMB) sessions, System Information, Temp Directories, Users and Groups, WdSupportLogs, CollectionSummaryReport.xls

6. On the user account page of Microsoft defender -> Associated alerts are shared. (security groups, user accounts belong to and threat hunt ID is not shown)
But SID, Department, Job title and MDI alerts are also showing

7. Which filter is a part of an Alert notification rule? -> Alert Severity

8. Which tab is present on the incident page when investigating a particular incident in Azure Defender -> Summary, Alerts, Devices, Users, Mailboxes, Investigations and Evidence and Response

9. You need to use the Kusto Query Language (KQL) to construct a query that identifies Azure Defender alerts -> where ProductName == "Azure Security Center"

10. You have created a query designed to identify a specific event on your domain controller. You need to create several similar queries because you have multiple domain controllers and want to keep each query separate. The solution should minimize administrative effort. Which three actions should you perform in sequence to clone a query?

- > On the Hunting page of Azure Sentinel, find the query you wish to clone.
- > Choose Clone query by clicking the ellipsis icon at the end of the row.
- > On the Create custom query page, make your edits, then click the Create button.

11. Azure Defender is for Cloud workload protection (main role of azure defender), not for Cloud Security posture, security, and configuration management.

12. Which selection helps you ensure Azure Defender is enabled over all the resources in a Subscription -> Automatic provisioning

13. Azure Arc -> extends Azure's management and governance capabilities to resources outside of Azure, such as on-premises servers, Kubernetes clusters, and even other cloud environments. It provides a unified platform to manage, secure, and monitor these resources as if they were running in Azure.

14. Which attribute of Azure Defender inspects registries and files of application software, operating system, and others for any changes that might point out an attack ---> File integrity monitoring

15. What are the extensions of auto-provisioning?

- Policy Add-on for Kubernetes
- Microsoft Dependency Agent
- Log Analytics agent for azure VMs

16. The threat intelligence report contains attacker information if available such as tools, tactics and procedures.

17. If you have a security recommendation that is not applicable for your environment, and you don't want to negatively affect your secure score, which option is the most appropriate to use --> Create an exemption for the recommendation.

18. When configuring GCP Connector in Azure Defender, which component is mandatory to have already configured in GCP --> GCP Security Command Center

19. Just-in-Time (JIT) VM access is a feature in Microsoft Defender for Cloud that helps protect your virtual machines (VMs) from unauthorized access by minimizing the time management ports are open. It works by locking down inbound traffic to your VMs and only allowing access when it's explicitly requested and approved. This reduces the attack surface while ensuring legitimate users can access the VMs when needed.

20. When reviewing Just-in-Time VM access, you noticed that some VMs appear under "Not Applicable". What are the reasons that must be present for a VM to be considered not applicable ->

JIT is not supported on VMs where:

It is missing a network security group (NSG) or Azure Firewall.

And Deployed as classic VM will not work --> JIT only supports VMs deployed through ARM (Azure Resource Manager), not 'classic deployment'.

21. Capabilities given below are part of Azure Defender for Servers

- A. Adaptive Application Control
- B. Integration with Qualys for Vulnerability Assessment
- C. Adaptive Network Hardening
- D. Fileless attack detection for Windows

And this is not a feature of azure defender.

E. Vulnerability assessment for Azure Container Registries --> (typically handled by Azure Defender for Containers)

22. Azure Sentinel for SAP, also known as Microsoft Sentinel solution for SAP (Systems, Applications, and Products) applications, is a security tool designed to protect SAP systems and applications from advanced threats. It provides continuous monitoring across all layers of SAP systems, including business logic, application, database, and operating system.

--> Azure Sentinel for SAP supports both cloud-based and on-premises implementations of SAP systems

23. In Azure Sentinel, a CI/CD pipeline (Continuous Integration/Continuous Deployment) is used to automate the deployment and management of custom content, such as analytics rules, automation rules, hunting queries, parsers, playbooks, and workbooks.

24 Which API should be used to assist with managing content through a CI/CD pipeline -> Azure Sentinel Management API

25. When a string is obfuscated, it means that the data within the string has been deliberately altered to make it difficult to understand or interpret. This is often done for security reasons, such as protecting sensitive information.

26. What does the "h" in front of a string literal such as 'h'my string' mean --> The string is obfuscated

27. Azure Sentinel requires explicit permissions to run playbooks (additional permissions). If a playbook shows unavailable, it means that Azure Sentinel does not have these permissions.

28. In queries "PCoHValue" means the ProcessCountOnHost value

29. ASIM stands for Advanced Security Information Model. It is a framework used in Microsoft Sentinel to normalize security data from various sources, making it easier to analyze and correlate information across different logs.

30. Which of the following are valid parsers in the ASIM -> Source-agnostic

ASIM includes two levels of parsers: source-agnostic and source-specific parsers.

31. You need to implement the Azure Information Protection requirements. What should you configure first -> Advanced features from Settings in Microsoft Defender Security Center

32. You can find the column names for the BehaviorAnalytics table in the Log Analytics workspace

33. To remediate active attacks you can use Azure Logic Apps.

34. To correlate data between two tenants in Microsoft Sentinel, you can use the workspace() expression along with the union operator.

The workspace() function allows you to reference tables from different workspaces.

To correlate data between two tenants in Microsoft Sentinel, you can use the workspace() expression along with the union operator.

35. When integrating cloud app security to Sentinel

Security extension should be added to Cloud app security portal
Data connector should be added to Azure Sentinel

- Azure Resource Manager (ARM) Templates are **JSON files** used to define and automate the deployment of **Azure resources** like VMs, networks, storage, etc.
- **Just-in-Time Access (JIT)** is a vital security protocol where permission to access applications or systems is only for a preset limited timeframe on an as-needed basis.
- Subscription Level (A): Enabling Azure Defender at the subscription level ensures that all resources within the subscription are covered. This includes VMs, network resources, and other Azure services that might benefit from JIT VM access and network security detections.

- **Live Response** is used in Microsoft Defender for Endpoint to investigate and take actions on devices, while **Live Stream** in Microsoft Sentinel allows real-time viewing of incoming log data for detection and analysis — they serve different purposes and are not the same.
- Livestream only works with **real-time data**, not historical data. So remove time limitations.
- In the context of Microsoft Sentinel and security data analysis, **"Delta values"** for hunting queries refer to the differences or changes in certain data points over time. These values are crucial for identifying anomalies, trends, and unusual activities that might indicate potential security threats.
- To calculate **delta values** in hunting queries, **hard-coded time filters like ago(7d) must be removed**; instead, use dynamic time settings from the hunting UI.
- Use a **livestream** in Microsoft Sentinel to test custom queries against real-time events, helping you detect matching log activity as it happens for rapid validation and investigation.
- **Bookmarks** help preserve important queries and results in Sentinel for threat hunting and collaboration.
- **Livestream** lets you test queries against live events and get notified of matches.
- **Watchlists** store external data lists to be referenced in queries.
- **Parsers** normalize diverse log fields into a consistent schema for easier querying.
- **Azure Monitor Workbooks** in Sentinel help create custom dashboards for visualizing and monitoring data.
- **Jupyter Notebooks** support ML, visualization, and analysis, ideal for investigations and threat hunting.
- **Hunting Query** is a predefined query used to explore potential threats in Sentinel.
- **Playbook** is an automation workflow built using Logic Apps in Sentinel.
- To visualize sign-in information over time in Microsoft Sentinel, use **Azure Monitor Workbooks**, which allow you to build interactive dashboards and reports from connected data like Microsoft Entra ID logs.
- **Custom network indicators**: Used to block or allow specific IPs, domains, or URLs via indicator lists.
- **Web content filtering**: Blocks unwanted content and monitors web activity; enforced via web filtering policies.
- **EDR block mode**: Enables Defender for Endpoint to stop malicious actions using behavior-based detection.
- **Scoped device group correlation**: Limits alert correlation to specific device groups; by default, correlation spans the full tenant.
- By default, Microsoft Defender for Endpoint retains data for **180 days**, but you can customize the retention period between **30 to 180 days** to meet compliance and regulatory needs.
- Threat analytics email notification rules send reports on new or updated threat insights, while Advanced Hunting detection rules can alert users to new or updated vulnerabilities using custom queries.
- To onboard non-Azure VMs like those in AWS for Defender for Cloud, you must install the **Azure Connected Machine agent** for the AWS server, which enables log collection and security assessments.
- In Microsoft Defender for Cloud, **data collection** provides visibility into security and health status of compute resources. **Defender plans** define protection levels for resource types. **Auto provisioning** automatically installs security extensions on machines, while **workflow automation** uses Logic Apps to respond to security alerts.
- Microsoft Defender for Cloud supports hybrid environments, including on-premises, AWS, and GCP—but requires deploying a connector for AWS or GCP integration.

- Microsoft Sentinel offers role-based access like Responder for viewing data, Contributor for editing resources, Logic App roles for managing playbooks, and Automation Contributor for automation rule integration—each role with specific permissions.
- **Azure Lighthouse** enables scalable, secure, and automated **multitenant management**, allowing service providers and enterprises to manage resources across multiple tenants while customers retain control over access and actions.
- To create or delete Microsoft Sentinel workbooks, a user must have the **Microsoft Sentinel Contributor** role (or lesser Sentinel role) **plus** the **Workbook Contributor (Azure Monitor)** role. This role is **not needed to view workbooks**, only for creating/deleting them.
- To ingest Syslog and CEF logs (common event format) into Microsoft Sentinel from devices where you can't install the Log Analytics agent directly, first deploy a **Linux machine** to act as a collector and forward logs to your Sentinel workspace.
- CEF logs are used to **standardize and streamline log ingestion** from diverse security products into platforms like Microsoft Sentinel for **centralized monitoring and analysis**.
- To connect a threat intelligence platform to Microsoft Sentinel, **register an app in Microsoft Entra ID** to get an Application ID and Client Secret, input them into your platform, and enable the Threat Intelligence Platforms data connector in Sentinel.
- **Microsoft Purview Insider Risk Management** helps detect data theft by departing users through activity indicators like file downloads or cloud transfers. Defender for Cloud Apps file policies enforce compliance and protect sensitive data. Identity Governance access reviews manage user access efficiently, and Office 365 alert policies trigger notifications based on predefined rules.
- The **Risky Users report** helps review and classify risky sign-ins as safe or compromised. **Risky Sign-ins report** shows accounts flagged as risky, while **Entra workbooks** assist in analyzing risks and detecting suspicious app or service principal activity.
- **Defender for Identity** uses on-prem AD signals to detect advanced threats and insider risks.
- **Microsoft Entra ID Protection** leverages Microsoft's threat intelligence to safeguard user identities.
- **Defender for Endpoint** is an enterprise endpoint security platform for advanced threat defense.
- **Defender for Office 365** offers cloud-based **email filtering** to protect against threats in collaboration tools.
- In Microsoft Sentinel, you can view incidents across **multiple workspaces** by selecting their checkboxes and clicking "**View incidents**". However, this **multi-workspace view is only available for incidents**
- Microsoft Sentinel supports **incident investigations up to 30 days old**. Incidents older than 30 days **cannot be reviewed or analyzed** directly in Sentinel.
- In Microsoft Sentinel, the **Entity Behavior** blade provides a detailed view of all activities and alerts related to a specific user. It aggregates alerts, anomalies, and timeline-based insights to help with comprehensive user investigation.
- In Microsoft Sentinel, automation rules can include conditions based on the **incident provider**, allowing you to specify whether the provider is Microsoft Sentinel, Microsoft 365 Defender, or both. This setting is independent of the data connectors in use and supports logical operators like *Equals* or *Does not equal*.
- In Microsoft Sentinel, playbooks can be manually triggered on both **incidents and alerts** for greater control. This is done from the *Incidents* page by selecting the incident, viewing full details, going to the *Alerts* tab, and then selecting the appropriate playbook under **View playbooks**.

- To run playbooks in Microsoft Sentinel, you must assign the **Microsoft Sentinel Automation Contributor** role to the **Azure Security Insights** identity on the playbook's resource group. Without this role, the playbook will not appear in Sentinel—even if other roles like Logic App Contributor are assigned.
- A **SecOps mailbox** collects **unfiltered messages** for analysis. Filters in EOP and Defender for Office 365 **take no action** on emails sent to this mailbox. **Quarantine policies** control user actions on quarantined emails. **Enhanced Filtering** ensures IP and sender info are preserved. The **Tenant Allow/Block List** lets admins override verdicts during **mail flow**, except for internal messages.
- **Defender for Cloud Apps** acts as a **Cloud Access Security Broker (CASB)**, enforcing security policies by brokering access between users and cloud resources in real time.
- **Defender for Office 365** offers **email filtering** to protect against email threats.
- **Defender for Endpoint** is an **endpoint protection platform** for threat prevention and investigation.
- **Defender for Containers** is a **cloud-native security solution** to protect and manage containers and their workloads.
- Alerts from Microsoft security solutions like Defender for Cloud Apps do **not automatically create incidents** in Microsoft Sentinel. They appear as raw data in the SecurityAlert table. To convert them into incidents, use a **Microsoft incident creation rule** under **Analytics > Create > Microsoft incident creation rule**.
- You can review the query logic of **near-real-time (NRT) rules** and **scheduled rules**, as they are not based on proprietary ML algorithms. However, **Fusion rules** and **ML behavior analytics rules** use Microsoft's internal machine learning models, so their query logic cannot be viewed or modified.
- In **Microsoft incident creation rules**, you choose the product (e.g., Defender for Cloud) for incident generation from alerts (**it's in the Analytics rule logic setting**). However, **query logic, alert details, and entity mapping are not available** in these rules — they are only available in **scheduled analytics rules**.
- To build ASIM (Advance security information model) parsers in Microsoft Sentinel, you must **collect representative logs**, connect the source system, and map event fields to the ASIM schema.
- Parser performance in Microsoft Sentinel can be improved by **pre-filtering** data before parsing **using filtering parameters**. This reduces post-processing overhead. Adding watchlists, time filters, or configuring for specific sources does **not** enhance parser performance.
- In Microsoft Sentinel, incident investigations require **analytics rules with entity mapping**. Since **entity mapping is only supported in scheduled and near-real-time (NRT) rules**, you can only investigate incidents generated by **Rule1 (NRT)** and **Rule3 (Scheduled)**.
- To collect logs and resolve vulnerabilities on a non-Azure VM (like one hosted in AWS), you should install the **Azure Connected Machine agent** to AWS server. It enables **Microsoft Defender for Cloud** to monitor and protect hybrid machines outside Azure.
- Microsoft Defender for Cloud delivers advanced threat protection and security alerts for a variety of Azure resources such as **VMs, Azure SQL, containers, App Services, storage, and key vaults**. When enabled, it protects resources like **VM1, storage1, and App1** under the respective Defender plans.
- Microsoft Defender for Cloud allows exporting detailed security alerts and recommendations to tools like **Log Analytics, Azure Event Hubs**, or external **SIEM/SOAR** platforms. With **continuous export**, you can stream real-time data or schedule regular exports, fully customizing what is sent and where.

#	Azure Security Solution	What It Protects
1	Microsoft Defender for Cloud	VMs, storage, SQL databases, containers, App Services, key vaults, DNS, hybrid cloud
2	Microsoft Defender for Endpoint	Endpoints (Windows, macOS, Linux), detects and responds to advanced threats
3	Microsoft Defender for Office 365	Emails, SharePoint, OneDrive, Teams – protects against phishing, malware, spoofing
4	Microsoft Defender for Identity	On-prem Active Directory – detects compromised credentials and insider threats
5	Microsoft Sentinel	SIEM/SOAR – collects, correlates, and responds to logs and incidents across platforms
6	Microsoft Entra ID Protection	Azure AD users – detects risky sign-ins, user behavior anomalies, and identity risks